

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester II Examination, 2023-24

Paper No. CHB-201: Inorganic & Physical Chemistry-I

Time: 3 Hours

Maximum Marks: 70

Instructions: Answer five questions in total, including Question 1 from each section which is compulsory. Use separate answer books for Section A and Section B.

SECTION A

(Inorganic Chemistry-I – Marks: 35)

Question 1. Answer all parts of the following:

[5 × 2 = 10]

- (a) Differentiate between a nitride and an azide.
- (b) What do you mean by sodide/potasside or alkalide ions in general?
- (c) Compare BH_3 and CH_4 in terms of total number of available orbitals and electron pairs in their valence shells.
- (d) Comment on the $2e^-$ - $3c$ bonds of diborane and write one preparation method.
- (e) Write the names of the hydrides of the nitrogen family and discuss their relative stability and reducing power.

Question 2.

- (a) Predict hybridization, draw structures, and explain shapes of: (i) XeF_4 and (ii) XeOF_4 .
[6]
- (b) Describe the structure of basic beryllium acetate. Explain why beryllium forms this basic acetate but other alkaline earth metals do not.
[6]

Question 3.

- (a) Discuss the interstitial and covalent carbides, giving their preparations and reactions.
[6]
- (b) Discuss graphite intercalation compounds (GICs). Why is graphite conductive whereas diamond is an insulator? ...
[6]

Question 4.

- (a) State the Kapustinskii equation for lattice energy and discuss how it is useful for calculating the lattice energy of crystals where ionic radii are not exactly known. [6]
- (b) Discuss the structure of crown ethers and cryptands. Explain how they are used for alkali metal complexation. [6]

SECTION B

(Physical Chemistry-I – Marks: 35)

Question 5. Answer all parts of the following: [5 × 2 = 10]

- (a) What is the unit of R (universal gas constant) in SI and CGS units?
- (b) Arrange in decreasing order of magnitude: root-mean-square velocity, average velocity, and most probable velocity.
- (c) Explain the Joule-Thomson experiment and find the value of the Joule-Thomson coefficient (μ_{JT}) for an ideal gas.
- (d) Define and derive the expression for critical temperature (T_c) of a van der Waals gas.
- (e) Justify the statement: “To maintain a small bubble needs excess pressure than a bigger one.”

Question 6.

- (a) Explain the statement: “A gas can be liquefied at $T < T_c$ and $P > P_c$.” Discuss the liquefaction of gases. [6]
- (b) Sketch the Carnot cycle in an S - V (Entropy-Volume) diagram, labelling all the states involved. [6]

Question 7.

- (a) What is a pseudo-first-order reaction? Give one example and write the expression for its rate constant. [6]
- (b) Derive the rate equation for a second-order reaction of the type $2A \rightarrow \text{Products}$. Calculate its half-life period. [6]

Question 8.

- (a) Define surface tension. Discuss the drop-weight method for the determination of the surface tension of a liquid. [6]
- (b) Write short notes on: [6]
 - (i) Viscosity and its temperature dependence
 - (ii) Maxwell-Boltzmann distribution of molecular speeds

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